

Faculty of Information Technology**Semester 2, 2025****FIT5145: Foundations of Data Science****Assignment 2: Description****Due Date: Friday, Week 8 (Sep 19, 2025), 11:55 PM**

The aim of this assignment is to investigate and visualise data using **R**. It will test your ability to:

1. Read data files and extract related data from those files;
2. Clean and process the data to ensure the results are valid.;
3. Perform exploratory data analysis and visualisation;
4. Use basic tools for managing and processing data; and
5. Communicate your findings in your report.

Assessment Details:

- Assessment Type: Individual Assignment
- Total marks: 15%
- Due Date: Friday, Week 8 (Sep 19, 2025), 11:55 PM. Please notice that we do not accept submissions after Sep 26, 2025 (i.e., 7 days after the due date).

Submission Details:

You are required to submit two separate files (**an RMD file and a PDF report**), both containing identical content.

1. The **RMarkdown** file includes your *(a) code, (b) answer, and (c) explanation* for each question.
(a) code: For better readability, please use separate code chunks for each question or task.
(b) answer: Please make sure to include the code outputs and written answers for each question.
(c) explanation: Please explain how you answered each question (i.e.: explaining your codes or summarising your work for each question).
2. A **report in PDF**: Please convert (knit) the RMD file to HTML and print the HTML as a PDF. This means that the contents of the RMD and the PDF are identical.

Please do NOT zip your submission files. Zip file submission will have a penalty of 20% of the total mark of the assignment. Failure to submit any of the required files will result in a loss of marks associated with the evaluation of those files. If you submit an incorrect version of a required file, you may request a resubmission; however, a late submission penalty will apply to the entire assignment to ensure marking fairness and consistency. Please ensure that you submit the correct versions of all required files!

Notes:

1. Whenever you find anomalies, errors, or any other issues in the dataset that negatively affect the answers, please determine the appropriate approach and perform the necessary data wrangling. This assignment also tests your ability to identify any issues in the data and preprocess them effectively for analysis.
2. Whenever a question asks for a certain value, your code should produce the value. For example, when a question asks for the number of rows contained in a table, your code should print out the number of the rows. Extraction of the answer manually by eye-examination will not earn any marks.
3. **Please check whether you can select and highlight texts in your PDF**, as shown below. This ensures that the turnitin score can be generated properly for your PDF file (we just need the Turnitin score for the PDF file, not the RMD file).

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1. Read data files and extract related data from those files;
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Assignment Task:

The dataset for this assignment is in the file *World_development_Indicators.csv* and it can be accessed from the Assessments page on Moodle.

The world bank produces a large dataset of development indicators on all parts of the world. A sample of this data is presented here in a .csv file for analysis. The data in the file represents readings of some development indicators (i.e., the column “*Series Name*” in the data file) for many countries in years from 1960 to 2019. Each row in the table contains the values of a certain series for a certain country across different years. For instance, a sample row in the data is shown below, it details the adolescent fertility rate in Argentina from 1960 to 2019. Please note that some values are missing in the data file.

Country Name	Country Code	Series Name	Series Code	1960	...	2019
Argentina	ARG	Adolescent fertility rate (births per 1,000 women ages 15-19)	SP.ADO.TFRT	61.39	...	62.35
...	...	...	...	...	...	...

Imagine you are working as a data scientist at a consulting company. Your team leader has asked you to analyse the dataset to generate insights based on the following questions. For this assignment, you are required to perform data analysis using the *World_Development_Indicators.csv* file.

1. How many unique values are there in the Series Name column of the data file? (6 marks)
2. How many comments in the Trend Description column describe a decreasing trend with positive feedback? (5 marks)
3. We want to explore the GDP series data, i.e., values with the Series Name 'GDP (current US\$)'. Specifically, instead of considering all countries, we will focus on: (1) the top five countries and (2) the bottom five countries based on their GDP (current US\$) values in 2018. Please present these 10 countries, along with their GDP values, in a single table and create a suitable chart. (4 marks)
4. Use a single R function or command to display the statistical information (i.e., minimum, maximum, and mean) of GDP values for each year from 2010 to 2019 for the 10 countries identified in Question 3 (Note: You may use multiple functions/commands to prepare the pre-processed data table, but when you compute and display the statistical information, you need to use a single R function/command.) (3 marks)
5. In which year did China record its maximum GDP? Which country shows the most largest increase in CO2 emissions over years? (3 marks).
6. Let's investigate life expectancy, fertility rate and GDP;
 - A. Identify the top 5 countries with the strongest relationship between life expectancy and GDP values across years. Please display both the countries and the corresponding values used to measure the strength of the relationship. (3 marks)

- B. Determine the top 5 countries that exhibit the most significant decreasing trend in fertility rate since 2010. Please display both the countries and the corresponding values used to quantify the decreasing trend. (3 marks)
7. The population density of countries has varied over time. Please divide the years into the following 6 periods and label them as Period 1-6, respectively, as shown below:
- i. 1960 - 1969 : Period 1
 - ii. 1970 - 1979 : Period 2
 - iii.
 - iv. 2010 - 2019 : Period 6

After that, write the code to compute the average population density value for each country within each period and display the first 10 data records of the output. Please make sure you sort the average population density value in a descending order for the output. (3 marks)

8. Create a chart that illustrates the trend of GDP data (i.e. GDP (current US\$)) for the USA. Identify and address any issues in the data, and justify your approach in resolving them. (4 marks)
9. Using both the original datasets and external sources, investigate the factors influencing the yearly trend of GDP (current US\$) values in selected countries. You may choose one or more countries for this investigation and should analyse at least 20 years of GDP data. Marks will be awarded based on the depth of your analysis (including discussion) and the strength of your reasoning (i.e., how insights and conclusions are drawn). Note: Please provide source links for any external data used. (7 marks).